

Learning That Remembers You: An AI-Native Ecosystem for Adaptive, Memory-Aware, and Multimodal Education at Scale

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<https://github.com/Dhanikesh-Karunanithi/Sudar>

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Abstract

Traditional learning management systems (LMSs) deliver static, one-size-fits-all content with no longitudinal learner model and no adaptive tutoring. Intelligent tutoring systems (ITS) that do adapt remain either narrow-domain research prototypes or products disconnected from the course-hosting infrastructure most organisations already use. We present **Sudar**, an AI-native learning system with three main contributions: (1) a full open-source reference platform that unifies authoring, delivery, and intelligence around a persistent **Digital Learner Twin**, adaptive sequencing, a *modality-agnostic* delivery architecture (read/listen, AI-generated maps and cards, optional watch/podcast/play, and SCORM) with course-dependent availability, plus recent product layers for consent-gated generative personalization, learner engagement (gamification telemetry), and procurement-oriented trust documentation; **SudarAgents**, a bounded tool-orchestration layer with an auditable `agent_runs` trail for repeatable micro-missions (e.g. learner week-plan sketches, cohort path diagnostics) grounded in the same Twin and telemetry as next-best-action; and an AI tutor with longitudinal cross-session memory and proactive *tap-to-reply* nudges; (2) the **Adaptive Learning Layer (ALP)**, an architecture and reference HTTP API by which host LMSs can forward events and call tutor and next-action services—reducing full platform replacement while making learner memory and adaptation portable; and (3) an **observed marginal AI cost structure** (open-weight inference and low-cost TTS) that we report *separately* from total cost of ownership for commercial LMS licences, so readers can compare like-with-like (API inference versus platform fees). The reference implementation is open source under the Apache-2.0 licence [11], evidence-informed by the learning sciences, and designed as an extensible bedrock platform on which the community

can build additional modalities, intelligence plugins, and LMS connectors.

1 Introduction

Learning management systems used by most organisations and universities deliver one-size-fits-all content: the same course for every learner, no memory of who the learner is, and no adaptation of sequence, difficulty, or support based on behaviour or prior knowledge. Research consistently shows that adaptive instruction and intelligent tutoring outperform fixed instruction [22, 2], yet mainstream LMS products do not maintain a longitudinal learner model or provide personalised, memory-aware tutoring [23, 1]. ITS that do adapt are typically not integrated into the same platform that hosts courses, compliance paths, and certifications.

We address this gap with Sudar, an AI-native learning system that (1) implements a complete reference platform with learner memory, adaptive paths, and multimodal delivery with explicit maturity tiers (core text and audio generation; optional studio-authored video and podcast; SCORM import and delivery); (2) is designed so its core components can be deployed as a layer on top of existing LMSs—the **Adaptive Learning Layer (ALP)**—enabling incumbent systems to gain longitudinal learner modelling, memory-aware tutoring, and modality choice without full platform replacement; and (3) demonstrates that this entire capability set can be delivered at low *marginal* AI infrastructure cost, reducing one major barrier that has historically made AI-native learning expensive at scale (while acknowledging hosting, compliance, content work, and connectivity remain real costs).

The economic dimension deserves particular attention. In 2026, AI inference costs have fallen approximately 1,000-fold since 2022 [4]. Delivering a full

personalised learning session with AI tutoring, content generation, and TTS narration now costs less than a fraction of a US cent on open-weight model providers. Yet commercial LMS vendors continue to charge \$5–\$20 per user per month, with AI features offered as credits-based add-ons requiring enterprise contracts with annual commitments of \$30,000 or more. This pricing structure excludes the very institutions—community colleges, NGOs, corporate L&D teams at mid-market companies, and educational organisations in the Global South—that would benefit most from adaptive, personalised learning. Sudar is an explicit response to this gap.

This paper presents a *system and architecture contribution* with an embedded economic analysis. Empirical efficacy data from a controlled study are not yet available; a pilot study has not yet been initiated; however, implementation is planned in collaboration with educational, university, and corporate institutions currently under negotiation, and corresponding outcomes will be documented in a subsequent publication. Our claims here concern design, implementation completeness, economic evidence, and the evidence-informed rationale for each component.

2 Background and Related Work

Adaptive instruction and ITS. Adaptive learning systems that tailor content and difficulty to the learner improve outcomes compared to one-size-fits-all instruction [22, 2]. Persistent learner models allow systems to adapt over time [18, 6]. Sudar implements adaptation through learner profiles, next-best-action recommendations, and adaptive path ordering.

Multimodal learning and dual coding. Dual coding and multimodal presentation support different learners and deepen encoding [14, 7]. Content is authored once and can be delivered in multiple formats depending on course configuration (text with read-along narration, optional animated video and dialogue podcast, interactive mindmap, flashcards, optional game-style activities, and SCORM packages), with modality-aware delivery; **next-best-action** and twin rollups are implemented primarily in the Learn application today, with Intelligence providing tutor and generation endpoints (the Intelligence `/api/modality/recommend` route is not authoritative until wired to learner signals).

Self-regulated learning and metacognition. Self-regulated learning and metacognitive scaffolding

improve persistence and transfer [24]. Sudar supports this through progress visibility (streaks, completion rates), next-best-action surfacing, deadline awareness, and the tutor’s “Your context” panel—a live view of what Sudar knows about the learner’s goals and preferences—enabling learners to reflect on and correct their own model. Where organisations enable adaptive courses, optional per-enrollment generative overlays (e.g. role explanation) are consent-gated and stored separately from canonical module content.

Structured curricula and prerequisite ordering. Structured curricula and prerequisite sequencing help learners build knowledge coherently [8]. Sudar supports learning paths with mandatory and optional courses, unlock rules, and adaptive reordering of optional content by the Intelligence engine.

Formative assessment and retrieval practice. Retrieval practice and formative assessment with feedback improve long-term retention [17, 5]. Sudar uses in-module quizzes with immediate feedback and low-stakes struggle detection that feeds directly into the learner model. Five quiz mode archetypes—standard, predict-then-learn, confidence-tagged, scenario-fork, and peer-contrast—are implemented to vary the retrieval modality and reduce habituation effects.

AI in higher education and LMS. Reviews of AI in HE note a persistent gap between ITS research and mainstream LMS adoption [23]. Recent work on AI-augmented textbooks—e.g., *Learn Your Way* [12]—transforms source material with personalisation (grade level, interests), multiple views (immersive text, slides, audio, mind maps), and formative assessment; a randomised trial showed gains over a digital reader. However, *Learn Your Way* focuses on augmenting a single textbook or chapter within one product; it does not claim a longitudinal learner model across sessions and courses, nor does it provide an architecture to augment existing LMSs. Table 1 contrasts our system with *Learn Your Way* and a typical LMS+AI add-on.

Proactive and predictive AI tutoring. A growing body of 2025–2026 research focuses on AI tutors that move beyond reactive Q&A toward predictive intervention. *AgentTutor* [13] demonstrated multi-turn personalised teaching with dynamic strategy adjustment using curriculum decomposition and knowledge memory modules. Reinforcement learning-driven ITS literature suggests improved intervention timing and

Table 1: Contrast: Sudar + ALP vs. *Learn Your Way* and a typical LMS with AI add-on.

	Sudar + ALP	Learn Your Way	Typical LMS + AI
Scope	Full platform + plugin layer	Single-product reader	Course host + chatbot
Learner model	Longitudinal (Digital Twin)	Per-session / material	None or stateless
Tutor memory	Cross-session, cross-course	Not claimed	Stateless
Modalities	Read/Listen, optional video/podcast, map/cards, SCORM	Text, slides, audio, mind map	Usually text/video only
Authoring	Integrated (Studio)	N/A (ingests material)	Separate tools
Augment LMS	Yes (ALP plugins)	No	N/A
Open source	Yes (Apache-2.0)	No	Rarely
Cost (1,000 learners/mo)	~\$21	N/A	\$3,400–\$44,000

adaptation by learning policies from behavioural data rather than relying only on hand-coded rules. Sudar’s Intelligence layer is designed as a foundation for this trajectory.

Inference cost economics and accessibility. The cost of AI inference has fallen approximately 1,000-fold from 2022 to 2026, at a rate of roughly $10\times$ per year—faster than Moore’s Law and faster than the bandwidth cost curves observed during the internet expansion of the early 2000s [4]. This collapse makes AI-native learning economically viable at population scale for the first time, yet no major LMS vendor has restructured its pricing model to reflect it.

3 System Design

3.1 Platform Architecture

The system has three surfaces sharing one data layer:

1. **Studio** (admin/creator): course and path creation, AI-powered generation from documents or

URLs, analytics, compliance management.

2. **Learn** (learner): personalised dashboard, course viewer with modality switching, AI tutor sidebar (*Sudar*), progress tracking, and certificates.
3. **Intelligence** (backend): adaptive engine, AI tutor engine, modality dispatcher, next-best-action computation, learning-event processing, and TTS audio generation.

All three surfaces share a single source of truth (Supabase / PostgreSQL) for authentication, learner profiles, content, and events. Studio and Learn are Next.js 15 applications; Intelligence is a Python 3.11+ FastAPI service. Learner actions and tutor exchanges write to `learning_events` and `ai_interactions`; the Intelligence layer then asynchronously updates the learner profile and next-best-action.

3.2 Digital Learner Twin and Learner Model

A persistent **Digital Learner Twin** is stored in `learner_profiles`. It includes: modality affinity scores (per format, 0.0–1.0), learning pace and difficulty comfort, behavioural signals (session duration, completion rate, streak), a `preferences` field (JSONB) holding TTS voice selection, response length preference (concise / one-line / detailed), and modality preference notes, and an `ai_tutor_context` JSON field holding known concepts, struggles, learning-style notes, goals, and interaction summaries. This context is updated asynchronously from tutor interactions and quiz outcomes. The design aligns with open learner model principles [6] and supports personalisation across courses and sessions. Crucially, the learner can inspect and correct their own model at any time via the tutor’s “Your context” panel, supporting metacognitive engagement [24].

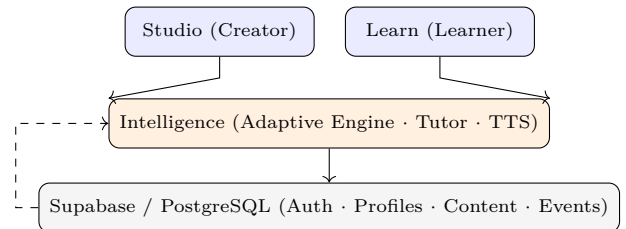


Figure 1: Sudar reference platform: three surfaces (Studio, Learn, Intelligence) over a shared Supabase data layer. Solid arrows show write/query flow; dashed arrow shows async profile update.

3.3 AI Tutor with Longitudinal Memory

The AI tutor, *Sudar*, is both *reactive* (retrieval-augmented generation over course content to answer learner questions) and *proactive* (generating nudges when struggle or a prolonged pause is inferred from behavioural signals). Every exchange is logged in `ai_interactions`. The tutor’s system prompt includes the learner’s memory summary drawn from `ai_tutor_context`, so responses reference prior struggles, match the learner’s preferred explanation style, and explicitly connect new material to prior learning. Most AI in LMS implementations are stateless [23]; Sudar is not.

Additional tutor capabilities include: a **text-selection popup** that lets learners highlight any passage and invoke one of six pre-built actions (“Explain this”, “Give me an example”, “Why does this matter?”, “Simplify this”, “Summarise”, “How does this connect...”) or a custom question; a **resizable overlay panel**; **proactive nudges** on the dashboard and in the course viewer with optional *tap-to-reply* chips that forward to the same tutor query pipeline; and a **generative block renderer** that allows the tutor to respond with structured artefacts—quiz questions, course recommendation cards, multi-step workflow trackers, and action groups—enriching interaction beyond simple Q&A.

3.4 Modality-Agnostic Delivery

Content is authored once. The delivery layer exposes a unified course viewer with a modality switcher; *which tabs are active depends on course configuration* (e.g. video scenes and podcast dialogue are optional fields in course settings). Learners always have Read, Listen, Watch, Map, and Cards; Podcast appears when configured; Watch is only fully active when the course includes generated or imported video scenes; an optional **Play** link appears when a module provides a SudarPlay asset. SCORM modules bypass the tab switcher and launch a full-viewport package player.

Text with Read-Along. A server-side TTS read-along path calls `/api/ai/generate-audio` per sentence, receiving neural voice audio (Edge-TTS by default). Each sentence is highlighted in-place as it plays, via `ReadingBodyWithSentences`. Telemetry includes `read_along_start` / `read_along_complete` events.

Listen (standalone audio TTS). A dedicated Listen tab (`AudioCard`) offers on-demand full-module TTS with optional regeneration and learner voice

preferences.

Animated Video (Watch). When `video_scenes` are present, the viewer renders scene-by-scene in-browser animation (kinetic, word-reveal, fade, list). Richer video can be produced via the separate SudarVid microservice and proxied by configuration. If no video is available, the Watch tab remains disabled.

Audio Podcast. When `podcast_dialogue` is configured, the viewer presents a two-voice dialogue through `DialogueSegment[]` with transcript and seek-by-segment playback.

MindMap (SudarMind). Generates an interactive concept map on demand via `POST /api/ai/generate-mindmap` using a custom SVG layout engine (Bézier edges, scope: module or course).

Flashcards. On-demand cards via `POST /api/ai/generate-flashcards`, supporting retrieval practice [17, 5].

SCORM. Packages render in a sandboxed iframe with a `postMessage` shim; scores and time write to `learning_events`. Extracted text can feed RAG for tutor grounding.

Modality-aware delivery and learner affinity scores inform UX in Learn; dual-coding principles [14, 7] motivate the design even where automation is partial.

3.5 Adaptive Sequencing and Next-Best Action

Learning paths support mandatory and optional courses and unlock rules. For optional courses, the ordering is adapted by the Intelligence engine from the learner profile and inferred knowledge gaps. A next-best-action algorithm suggests the next course or module with a human-readable explanation, surfacing required items and due-soon deadlines. Formative assessment and struggle detection feed the learner model [5, 17].

Five quiz mode archetypes—standard closed-response, predict-then-learn, confidence-tagged, scenario-fork, and peer-contrast—vary the retrieval modality and support metacognitive calibration.

3.6 Agent orchestration and auditability

Beyond conversational tutoring, Sudar exposes a gateway for *bounded* plans with a restricted tool catalogue: reads and scoped writes reuse the Twin, `learning_events`, and rollups, and each completed mission records structured output plus tool traces in `agent_runs`. This is a design/implementation

contribution: it operationalises recommendation and light reflection loops aligned with formative practice and calibration themes already emphasised above, without claims of new empirical efficacy relative to unstructured chat. Org policy can disable individual capabilities; integrators seeking a concise HTTP sketch can consult `docs/ALP_API.md` and `docs/AGENTS_PLATFORM.md`.

3.7 Plugin Architecture for Existing LMSs: The Adaptive Learning Layer (ALP)

The most novel architectural contribution is the **Adaptive Learning Layer (ALP)**: a plugin layer that packages Sudar’s capabilities as independently deployable services for existing LMSs. Organisations running Moodle, Canvas, Blackboard, or a proprietary LMS need not abandon their investment to gain learner memory and adaptive tutoring—ALP attaches these capabilities via APIs and event hooks. Moodle 4.5 introduced a dedicated AI subsystem with a formal separation between Provider plugins and Placement plugins [15]; ALP is architecturally aligned with this model.

ALP reference API and roadmap plugin surfaces.

1. **SudarMemory (reference).** The foundational layer. Host LMS sends xAPI, SCORM outcomes, or webhooks to Sudar Learn’s `/api/alp/events`; Sudar maintains a Digital Learner Twin per user. Vendor-packaged connectors are roadmap; the HTTP contract is implemented on Learn today.
2. **SudarChat (reference).** Embeds the Sudar AI tutor via Learn’s `/api/alp/tutor/query` and signed `/alp/embed` flows; Twin data is read from Supabase after org-scoped auth.
3. **SudarStudio Embed (roadmap).** Target: “Generate with AI” inside the host LMS editor; today generation is authored in Sudar Studio/Learn rather than as installable LMS editor modules.
4. **SudarRecommend (reference).** Next-best-action for the LMS dashboard via Learn’s `/api/alp/next-action`; canonical scoring lives in Learn’s `/api/intelligence/next-action`. No schema changes are required on the host database.
5. **SudarAdapt (roadmap).** Target: LMS conditional activity integration for unlock/reorder; Sudar Learn already supports adaptive path ordering for Sudar-hosted paths.
6. **SudarAgents (reference).** HTTP descriptors on Sudar Intelligence (`/api/agents/skills`, `/api/agents/alp-openapi.json`) document the bounded orchestration surface for partner backends; see `docs/AGENTS_PLATFORM.md`.

Integration flow. Figure 2 illustrates the integration. The host LMS continues to own content, enrolments, and grading. ALP sits alongside it, enriching the experience without replacing infrastructure. No data migration is required.

Reference implementation status. The repository provides authenticated HTTP endpoints on the Learn application (e.g. `/api/alp/events`, `/api/alp/tutor/query`, `/api/alp/next-action`, embed-token helpers) with behaviour and field mappings documented in `docs/ALP_API.md`. Host LMSs can call these without hosting Sudar’s UI. Packaging the same contract as distributable installable modules per vendor (Moodle, Canvas, etc.), hardening for multi-tenant production, and mapping institutional user identities, remain deployment engineering tasks rather than missing core APIs.

Addressable scale. Moodle alone has over 500 million registered users across 233 countries and holds approximately 28% of the global LMS market [16]. ALP connectors for Moodle, Canvas, Blackboard, and Sakai collectively address the largest possible deployment channel for Sudar’s intelligence layer without requiring a single institution to change its LMS.

Positioning. We position ALP as an open *reference architecture* and HTTP API for memory-aware services alongside incumbent LMSs; related work includes xAPI/LRS ecosystems, enterprise AI add-ons, and adaptive extensions—we focus on an Apache-2.0 reference stack with explicit trust and consent surfaces rather than claiming exclusive novelty.

3.8 Open-Source Extensibility and the Modality Upgrade Path

A distinguishing property of Sudar is that its modality delivery stack is explicitly designed for community extension and backend substitution. The `VideoScene[]`, `DialogueSegment[]`, and

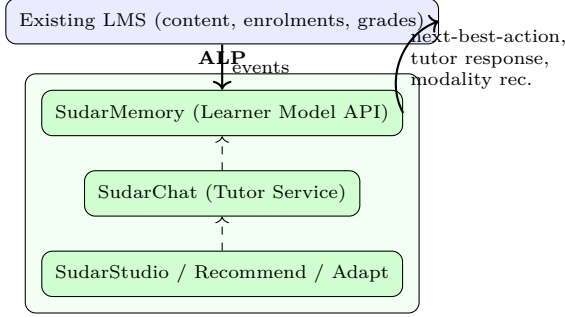


Figure 2: ALP integration architecture. The host LMS sends learning events to SudarMemory; ALP returns next-best-action, tutor responses, and modality recommendations. The LMS retains ownership of content and enrolments.

ModalityVariants types are provider-agnostic interfaces; any modality backend can be swapped without touching the delivery layer.

Video backend upgrade path. The current in-browser renderer delivers animated typographic video at zero infrastructure cost. Organisations requiring higher production value can connect any of the following backends to the same `VideoScene[]` interface: **Higgsfield** [9], a multi-model creative platform aggregating Sora 2, Kling 2.6, and Google Veo 3.1 with cinema-grade camera simulation; **Wan 2.1** [3] (Alibaba, MIT open-source), a text-to-video suite from 1.3B to 14B parameters enabling 480p generation on a consumer GPU; and **HunyuanVideo** [20] (Tencent, open-source), a Diffusion Transformer model suited for instructor avatar generation in corporate training contexts.

The principle is: Sudar is the authoring and delivery operating system; the generation backend is a plugin. A school in Sub-Saharan Africa self-hosts Wan 2.1 on a local GPU server at zero per-call cost. A Fortune 500 L&D team connects Higgsfield for cinematic course introductions. The learner experience—the Digital Learner Twin, adaptive sequencing, AI tutor, and modality switching—is identical.

Community extension patterns. The Apache-2.0 licence enables: new modality plugins (VR, AR, 3D simulation); new LLM backends (any OpenAI-compatible API endpoint via environment variable); new language and voice packs (Sarvam AI Indian English neural voices are already integrated alongside Edge-TTS); and vertical-specific forks for medical education, software engineering bootcamps, and regulated-industry compliance.

3.9 The Evolution of Sudar Intelligence

The Intelligence layer described above reflects the current implementation: reactive Q&A via RAG, rule-based proactive nudges (with structured follow-up *choices* for client surfaces), and a longitudinal memory updated from quiz outcomes and tutor exchanges. This is a substantial advance over stateless LMS AI; it is also, architecturally, only the beginning.

As the `learning_events` table accumulates interaction data, the Intelligence engine transitions from rule-based heuristics to *learned* intervention policies. The research frontier demonstrates what becomes possible: reinforcement learning-driven ITS approaches can improve intervention timing, adaptation, and learner persistence relative to static rule-only tutoring policies.

Sudar’s architecture anticipates this transition explicitly, with the Digital Learner Twin and `learning_events` stream serving as the shared substrate for future first-party and community-built Intelligence plugins: **SudarCoach** (career pathway advisor), **SudarAssess** (item response theory psychometric layer), **SudarSocial** (peer learning orchestrator), **SudarCompliance** (automated evidence collection for regulated industries), and **SudarTranslate** (real-time multilingual delivery).

4 Implementation and Reproducibility

The reference platform is implemented with Next.js 15 (Studio, Learn; TypeScript strict mode, App Router, Tailwind CSS), Python 3.11+ FastAPI (Intelligence; async handlers, Pydantic v2), and Supabase (PostgreSQL + pgvector, auth, storage). AI providers include Together AI (primary; open-weight models), OpenAI, and Anthropic. TTS uses Edge-TTS (default; zero cost, 40+ languages, 300+ voices) with optional Sarvam AI Indian English neural voices. The schema and event model are fully documented in the repository (`ECOSYSTEM.md`, `RESEARCH_FOUNDATION.md`).

Currently implemented: shared authentication and schema; course and learning path CRUD and publish; document-to-course and SCORM 1.2 import; learner dashboard with streak, time-on-task, Sudar recommendations, optional engagement surfaces (e.g. quests, coins, achievements) where enabled, and upcoming deadlines; multimodal course viewer as in Section 3.4 (read-along, Listen, optional Watch and Podcast, Map, Cards, optional

Play, SCORM); AI tutor with longitudinal memory, RAG, text-selection popup, resizable overlay, tap-to-reply proactive nudges, “Your context” live memory panel, and generative block rendering; five quiz mode archetypes and five lesson archetypes; next-best-action and adaptive path ordering; learning paths with unlock rules, shareable public certificate pages, browser print/save, and a server-generated PDF download route; optional per-enrollment generative overlays with org policy and consent; Studio governance and a technical trust pack for security reviews; compliance view (overdue / at-risk / on-track) and email reminders (cron); analytics rollups and risk signals (feature-flagged in production); time per section in analytics; global search; server-side sensitive-input guardrails on AI routes; and learner preferences. ALP Learn proxies are implemented (Section 3.7). Production deployment is documented: Studio and Learn on Vercel; Intelligence on Railway, Render, or Fly.io, with the Intelligence URL set in the apps. The project is open source under the Apache-2.0 licence [11].

5 Economic Accessibility: The Radical Cost Case

The economic dimension of Sudar is not incidental to its design—it is one of its three primary contributions.

Observed infrastructure cost. Operating Sudar at full engagement—10 courses generated, 20 learner review sessions with AI tutor chat, adaptive personalisation, mindmap generation, flashcard generation, and TTS narration—costs approximately **\$0.20 per week** on the Together AI + Edge-TTS stack. At organisational scale with 1,000 learners and 50 sessions per month, the contrast is:

Stack	Per month	Annual
Sudar (Together AI + Edge-TTS)	\$21	\$252
GPT-4o + OpenAI TTS-1	\$3,405	\$40,860
Claude 3.5 Sonnet + Azure TTS	\$3,735	\$44,820
Docebo (platform licence)	\$5,830+	\$69,960+

At 10,000 learners, Sudar’s open-weight stack costs approximately \$210/month. The delta versus a closed-model stack—over \$400,000 per year—is the economic value of open-weight inference returned to the institution rather than captured by AI API providers.

The inference cost collapse. AI inference costs have fallen approximately 1,000-fold from 2022 to 2026 at roughly 10× per year [4], driven by GPU efficiency gains, software optimisation (vLLM, TensorRT-LLM), Mixture-of-Experts architectures, and quantisation. GPT-3-class performance costs \$0.06 per million tokens today versus \$60 in 2021. The cost of delivering a world-class personalised tutor interaction to one learner is now less than \$0.02 per month—below the cost of a single SMS message. The economic argument against AI-native learning at scale has expired.

The education access crisis. As of 2024, 251 million children and youth remain out of school globally [21]. 2.6 billion people remain offline [10], with mobile internet in Africa costing 14× more than in Europe. An annual financing gap of \$39 billion exists for basic education in low- and lower-middle-income countries through 2030 [21]. The average corporate training spend per employee in the United States is \$1,283 per year [19]; Sudar’s *marginal* AI stack cost (\$0.21/user/month in our worksheet) is **not comparable** to that all-in training benchmark—we include it only to contrast order-of-magnitude *LMS+AI API* stacks in the table above, not to claim equivalent coverage.

High-end AI learning platforms priced at \$12–\$20 per user per month are not deployable in this context. A platform whose full AI infrastructure costs \$0.02 per learner per month—and whose stack is open-source, self-hostable, and runnable on open-weight models—is.

6 Evidence and Evaluation Strategy

Research foundation. The design is evidence-informed. RESEARCH_FOUNDATION.md maps each capability to learning-science references: adaptive instruction [22, 2], dual coding [14, 7], self-regulated learning [24], formative assessment [5, 17], intelligent tutoring [18, 22], and learner modelling [6, 18].

Scope of current claims. This is a system and architecture paper. Current claims are: (a) the platform implements a working reference deployment and the subsystems we describe, with *explicit* caveats for course-dependent modalities, optional product layers, and ongoing LMS packaging; (b) each design choice is grounded in established learning-science evidence; (c) the ALP architecture addresses a gap in

LMS-agnostic, memory-aware services that commercial stacks rarely ship as open, portable APIs; and (d) the economic cost structure has been empirically observed in practice and validated against current provider pricing.

Pilot study. A pilot evaluation is planned, to be initiated after the current build phase, and will be conducted in partnership with an organisational partner to assess adoption, engagement, and early learning outcomes. Results will be reported in a subsequent paper when available.

Future controlled studies. Plans include randomised or quasi-experimental comparisons of Sudar versus a standard LMS on learning outcomes, learner satisfaction, and time-to-competency, as well as studies isolating the contribution of the longitudinal tutor memory component and the cost-per-outcome metric versus incumbent platforms.

7 Discussion

Limitations. Reference ALP HTTP endpoints and Learn-side proxies are implemented, but *packaging* and operating them as one-click add-ons in each major LMS marketplace (and validating identity mapping in diverse SSO environments) still requires per-institution work. Controlled efficacy data are not yet available; claims about learning gains remain grounded in the underlying learning-science evidence rather than Sudar-specific trials. The Digital Learner Twin relies on inferred signals that may not fully capture a learner’s actual knowledge state [18, 6]. Edge-TTS uses Microsoft’s voice infrastructure under its terms; production deployments with paying customers should plan licensed TTS. The predictive capabilities in Section 3.9 require longitudinal data to learn reliably; live systems today combine rules with heuristics.

Future work. Planned work includes: first-party or community LMS distributables that wrap the same ALP contract; maturing analytics rollups and scheduled jobs at scale; SudarFeed (short-form social feed) and deeper SudarPlay integration; RL-style intervention policies; controlled efficacy and cost-per-gain studies per docs/PILOT_PLAN.md; SudarCoach, SudarAssess, and SudarCompliance-style intelligence plugins; enterprise features (SSO, HRIS); and an offline-first deployment path for low-connectivity settings.

8 Conclusion

We presented Sudar, an AI-native learning platform with three contributions: (1) a reference platform combining authoring, delivery, and intelligence around a Digital Learner Twin, adaptive sequencing, modality-agnostic delivery with course-dependent options, product layers for policy/consent, engagement telemetry, and trust documentation, and an AI tutor with longitudinal memory and proactive tap-to-reply nudges; (2) the Adaptive Learning Layer (ALP), an architecture and reference API for deploying these capabilities alongside existing LMSs—with the potential to reach the hundreds of millions of Moodle, Canvas, Blackboard, and Sakai users [16] without requiring LMS replacement; and (3) a demonstrated economic model that reduces the infrastructure cost of AI-native personalised learning to less than \$0.02 per learner per month, making it viable at the scale of national education systems and accessible in contexts where \$12–\$20/user/month platforms are structurally excluded.

The inference cost collapse is the macro condition that makes all of this possible [4]. Platforms built on open-weight models and open-source infrastructure are positioned to pass this economic gain directly to learners—particularly the 251 million children currently out of school [21] and the 2.6 billion people currently offline [10].

Together, these contributions offer a path to making adaptive, memory-aware learning accessible both through a standalone platform and as an augmentation of the LMSs that organisations already rely on. The reference implementation is open source, grounded in a broad learning-science evidence base [22, 2, 14, 7, 24, 8, 6, 5, 17], and designed as a bedrock upon which the community can build modalities, intelligence plugins, and LMS connectors.

Acknowledgments

The system and research are the author’s project. The reference implementation is available at <https://github.com/Dhanikesh-Karunanithi/Sudar> with documentation (ECOSYSTEM.md, RESEARCH_FOUNDATION.md, docs/sudar-memory.md).

Call for collaboration. Institutions, organisations, and open-source contributors are invited to collaborate on pilots, plugin integrations, and community extensions. Contact: connect@dhanikeshkarunanithi.com or <https://github.com/Dhanikesh-Karunanithi/Sudar>.

A Schema Summary

Key tables for reproducibility: `learner_profiles` (Digital Learner Twin: `ai_tutor_context`, `next_best_action`, modality affinity scores, behavioural signals, `preferences` JSONB); `learning_events` (`event_type`, payload, modality, duration); `ai_interactions` (`user_message`, `ai_response`, `context_used`); `enrollments` (path/course, progress, `due_date`, optional `personalization_overlays` JSONB); `certifications`. Full schema: `ECOSYSTEM.md` in the repository.

B Infrastructure Cost Reference

Stack		/learner/mo	Annual (1k)
Together AI 8B + Edge-TTS		\$0.021	\$252
Together AI 70B + Edge-TTS		\$0.105	\$1,260
GPT-4o + OpenAI TTS-1		\$3.41	\$40,860
Claude 3.5 + Azure TTS		\$3.74	\$44,820
Docebo (licence, low end)		\$5.83	\$69,960
Sana Labs (est., min. 300)		≈\$15	≈\$180k

AI infrastructure only; excludes hosting (Supabase free tier: \$0; Pro: \$25/month).

C Reproducibility and evaluation artefacts

Fill and version the cost assumptions in `docs/research/COST_WORKSHEET.md`; scope claims using `docs/research/EVALUATION_APPENDIX.md`; run pilots per `docs/research/PILOT_PROTOCOL.md`; extend `scripts/benchmark-sudar.mjs` for automated latency/token measurements when credentials are available.

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